SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 Total Marks:30

Annexure A

Code of Conduct:

I M.vishvanandh(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:M.vishvanandh

Question 1 – Breadth-First Search (BFS)

Graph: A → (B, C), B → (D, E), C → (F, G). Assume children are visited from left to right.

Iteration	Queue (front → rear)	Visited Node
1	[B, C]	A
2	[C, D, E]	B
3	[D, E, F, G]	C
4	[E, F, G]	D

5	[F, G]	E
6	[G]	F
7	[]	G

1. BFS traversal order: A, B, C, D, E, F, G.

2. Queue contents after each iteration are shown in the table. The queue is FIFO (First In, First Out).

Question 2 – Depth-First Search (DFS)

Using left-to-right traversal, DFS explores each branch as deeply as possible before backtracking.

Step	Node visited	Stack (top at right; simplified)
1	A	[A]
2	B	[A, B]
3	D	[A, B, D]
4	E	[A, B, E] after backtracking
5	C	[A, C] after backtracking
6	F	[A, C, F]
7	G	[A, C, G]

DFS traversal order: A, B, D, E, C, F, G.

Question 3 – 8-Puzzle: BFS

Initial state:

1 3 6
5 0 2
4 7 8

Goal state:

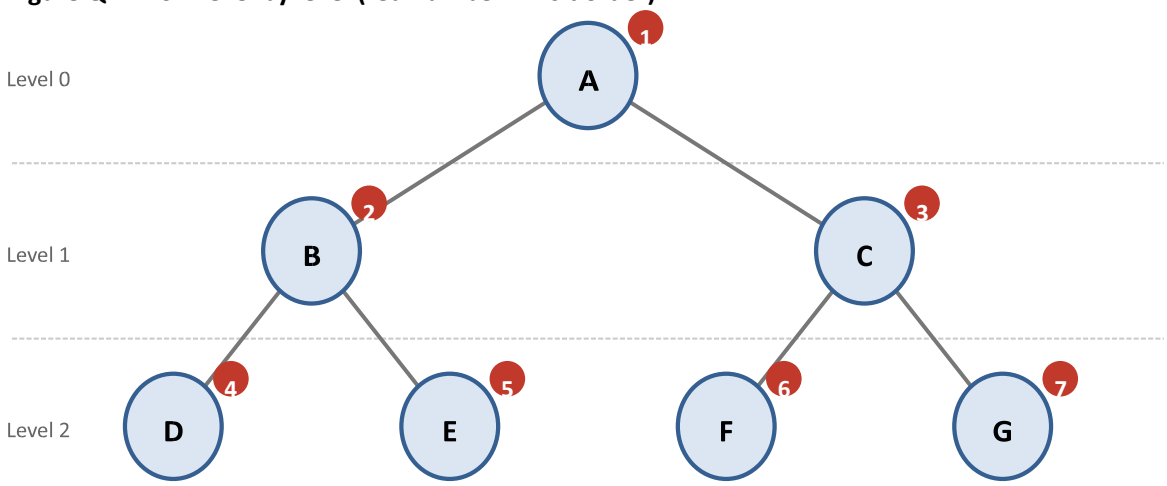
1 2 3
4 5 6
7 8 0

The five-row table in the question is not enough to reach the goal: the goal is reached at depth 8 (eight moves). The first five states removed from the BFS queue are listed below. Queue contents are abbreviated to avoid listing hundreds of states; BFS stores states in FIFO order.

Iteration	Current state expanded	Queue after expansion (summary)
1	1 3 6 / 5 0 2 / 4 7 8	New states from moving blank Up, Down, Left, Right (valid moves only)
2	1 3 6 / 5 2 0 / 4 7 8	Previously queued states

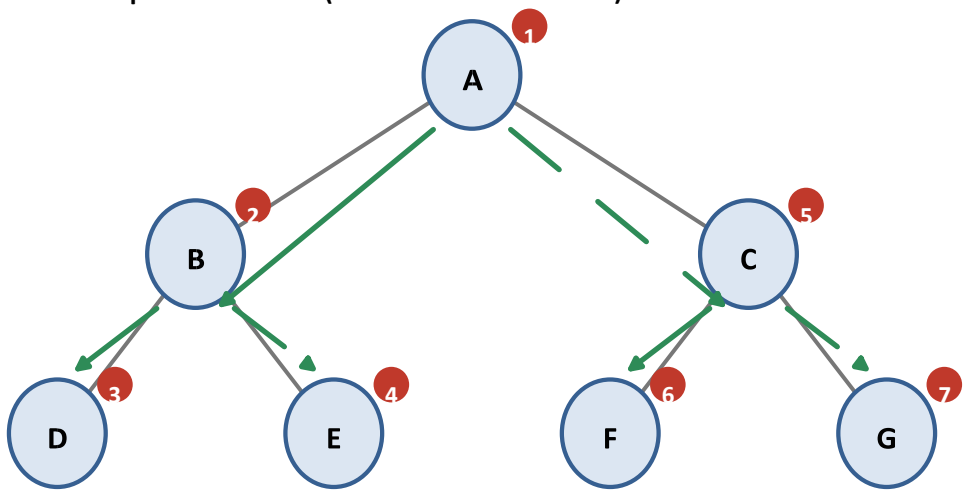
Supporting Diagrams – Questions 1 and 2

Figure Q1 BFS – level by level (red number = visit order)



Traversal order: A, B, C, D, E, F, G (queue is FIFO)

Figure Q2 DFS – deepest branch first (red number = visit order)



Traversal order: A, B, D, E, C, F, G (dashed arrow = after backtracking)

		remain first; newly generated states are appended
3	1 3 0 / 5 2 6 / 4 7 8	FIFO order maintained; repeated states are not added again
4	1 0 3 / 5 2 6 / 4 7 8	Unexpanded states remain in front of newly generated states
5	1 2 3 / 5 0 6 / 4 7 8	Successors are appended to the rear of the queue

1. Nodes expanded in the first five iterations are the five current states shown in the table.

2. The goal is first reached after expanding 311 states in this left/up/down/right neighbor ordering; 500 distinct states are discovered (including the initial state) before the goal is dequeued. Counts can vary if the order of legal moves is changed.

3. One shortest solution path (blank movement): Right → Up → Left → Down → Left → Down → Right → Right.

Initial: 1 3 6 / 5 0 2 / 4 7 8

Step 1: 1 3 6 / 5 2 0 / 4 7 8

Step 2: 1 3 0 / 5 2 6 / 4 7 8

Step 3: 1 0 3 / 5 2 6 / 4 7 8

Step 4: 1 2 3 / 5 0 6 / 4 7 8

Step 5: 1 2 3 / 0 5 6 / 4 7 8

Step 6: 1 2 3 / 4 5 6 / 0 7 8

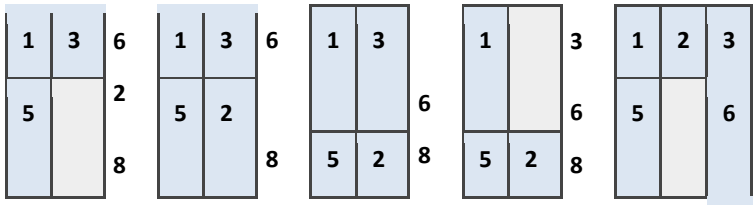
Step 7: 1 2 3 / 4 5 6 / 7 0 8

Goal: 1 2 3 / 4 5 6 / 7 8 0

4. BFS finds a shortest solution in an unweighted 8-puzzle because it explores states in increasing order of depth. Every move has equal cost, so the first goal state reached has the minimum number of moves.

Note: For the 8-puzzle, the original worksheet provides only five table rows, although reaching the goal requires more expansions. The full shortest path and explanation are included here.

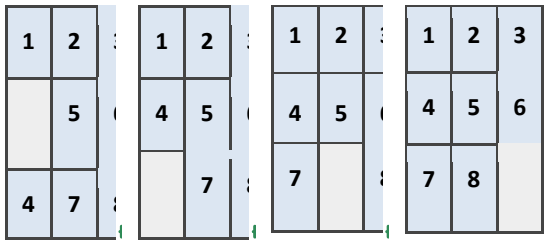
Figure Q3 Shortest solution path (blank moves shown under each board)





Initial Step 1 Step 2 Step 3 Step 4

Right Up Left Down



Step 5 Step 6 Step 7 Goal Left Down Right Right